MINHAJ UNIVERSITY LAHORE

School of Software Engineering

FYP Proposal Submission Form No 2

Project Title: (Max 2 lines)	TrustLens – AI-Powered Seller Trust Verification System for Online Marketplaces
Projects Members: (1 – 2 Only)	Name: Haseeb Ur Rehman Registration No: 2022F-mulbsswe-043 Program / Semester: B.S. Software Engineering / 8th Semester Signature: _____ Name: Muhammad Muttayab Registration No: 2022F-mulbsswe-012 Program / Semester: B.S. Software Engineering / 8th Semester Signature: _____
Supervised By:	Name: Mam Misbha Designation: _____ School of Software Engineering
Co-Supervisor:	Name: _____ Designation: _____ School of Software Engineering
Project Description: (Max 20 lines)	Objective / Purpose: TrustLens is a web application that verifies whether an online marketplace seller is genuine or fake by generating an explainable Trust Score, helping buyers avoid scams before making a deal. Scope: Sellers complete identity, phone, email, and device checks; the system analyzes each listing with AI and combines all signals into a single Trust Score (0–100) with a clear risk level and the reasons behind it. It also exposes a REST API so marketplaces can request a seller's score. Target Users: Online buyers, marketplace platforms (OLX-style), rental and property portals, second-hand vehicle sites, and freelance / peer-to-peer commerce platforms. Tools & Technologies: React.js / Next.js (frontend), Tailwind CSS (styling), Python with FastAPI (backend REST API and AI logic), PostgreSQL (database), and OpenAI / Claude API with OpenCV and EasyOCR (AI engine). Features: identity verification (CNIC OCR + face match), phone OTP and email checks, device and IP / VPN detection, AI listing analysis, an explainable Trust Score engine with risk levels, an admin and appeal panel, and a public Trust Score API. Expected Outcome: A working web application and API that return an explainable seller Trust Score within seconds, clearly showing why a seller is low, medium, or high risk.

	Future Scope: reverse-image stolen-photo detection, chat scam-detection, direct marketplace integrations, reputation history, and a mobile version.
Which real world problem shall be solved by this project?	Online marketplace buyers are frequently scammed by fake seller profiles, stolen photos, throwaway phone numbers and emails, multiple accounts created from one device, and unrealistic "too good to be true" listings, with no reliable way to judge a seller before paying. TrustLens solves this by verifying multiple independent signals — identity, phone, email, device, IP, account age, and AI listing analysis — and combining them into one explainable Trust Score. Buyers instantly see how trustworthy a seller is and why, which reduces fraud and increases confidence, while marketplaces can integrate the score through a simple API.
Development Related Information: **Tools** **DBMS** **Platform**	Tools: Visual Studio Code, Postman, Figma, Git & GitHub Frontend: React.js / Next.js, Tailwind CSS Backend: Python, FastAPI Database: PostgreSQL (with Redis cache) AI Integration: OpenAI / Claude API, OpenCV, EasyOCR (OCR), DeepFace (face match), and a custom-trained scam-listing classifier Deployment: Vercel (frontend), Railway (backend) Platform: Web-based, cross-platform (Windows / Linux / macOS browsers)
7th Semester:	Problem Identification Solution Proposal. SRS Document. **Proposal Presentation**
8th Semester:	Design Based Implementation. Testing/Validation. Project Report. **Final Presentation**
For Submission	1. **Supervisor Name**:_____ _____ **Supervisor' Signature** 2. **Muhammad Kamran** FYP Coordinator _____ **FYP Coordinator'** Signature 3. **Imran Akther**

Comments FYP Evaluation Committee **Form No 3**

	HOD School of Software Engineering _____ **HOD' Signature**

Project Title_____

Name of Committee Member 01_____

The decision by the FYP Evaluation Committee: (Please tick one option)	Approved		Date: _____
	Not Approved		
	Revise & Resubmit		
Comments FYP Evaluation Committee:			
Signature			

Name of Committee Member 02 _____

The decision by the FYP Evaluation Committee: (Please tick one option)	Approved		Date: _____
	Not Approved		
	Revise & Resubmit		
Comments FYP Evaluation Committee:			
Signature			

Muhammad Kamran FYP Coordinator/Lecturer School of Software Engineering	**Imran Akther** Head, School of Software Engineering